



Low-latency Perception and Action for Agile Vision-based Flight

Fact Sheet

Project Information

AGILEFLIGHT

Grant agreement ID: 864042

[Project website](#)

DOI

[10.3030/864042](https://doi.org/10.3030/864042)

Project closed

EC signature date

7 February 2020

Start date

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End date

31 August 2025

Funded under

EXCELLENT SCIENCE - European Research Council (ERC)

Total cost

€ 2 000 000,00

EU contribution

€ 2 000 000,00

Coordinated by

UNIVERSITÄT ZÜRICH
Switzerland

Project description

Towards autonomous drone navigation comparable to human pilots

Drones are still nowhere near capable of navigating through complex environments as well as human pilots. Obtaining more agile robots calls for faster sensors and low-latency processing. The EU-funded AGILEFLIGHT project intends to develop innovative scientific methods to demonstrate autonomous, vision-based, agile quadrotor navigation in unknown, GPS-denied and cluttered environments with possibly moving

obstacles. The aim is to achieve navigation that is as effective in terms of manoeuvrability and agility as that of professional drone pilots. To this end, the project will develop algorithms that combine standard camera and event camera advantages. AGILEFLIGHT will also develop novel methods that enable agile manoeuvres through cluttered, unknown and dynamic environments. This work could benefit disaster response, aerial delivery and inspection work.

Objective

Drones are disrupting industries, such as agriculture, package delivery, inspection, and search and rescue. However, they are still either controlled by a human pilot or heavily rely on GPS for navigating autonomously. The alternative to GPS are onboard sensors, such as cameras: from the raw data, a local 3D map of the environment is built, which is then used to plan a safe trajectory to the goal. While the underlying algorithms are well understood, we are still far from having autonomous drones that can navigate through complex environments as good as human pilots. State-of-the-art perception and control algorithms are mature but not robust: coping with unreliable state estimation, low-latency perception, real-time planning in dynamic environments, and tight coupling of perception and action under severe resource constraints are all still unsolved research problems. Another issue is that, because battery energy density is increasing at a very slow rate, drones need to navigate faster in order to accomplish more within their limited flight time. To obtain more agile robots, we need faster sensors and low-latency processing.

The goal of this project is to develop novel scientific methods that would allow me to demonstrate autonomous, vision-based, agile quadrotor navigation in unknown, GPS-denied, and cluttered environments with possibly moving obstacles, which can be as effective in terms of maneuverability and agility as those of professional drone pilots. The outcome would not only be beneficial for disaster response scenarios, but also for other scenarios, such as aerial delivery or inspection. To achieve this ambitious goal, I will first develop robust, low-latency, multimodal perception algorithms that combine the advantages of standard cameras with event cameras. Then, I will develop novel methods that unify perception and state estimation together with planning and control to

enable agile maneuvers through cluttered, unknown, and dynamic environments.

Fields of science (EuroSciVoc) ⓘ

[engineering and technology](#) > [electrical engineering](#), [electronic engineering](#), [information engineering](#) > [electronic engineering](#) > [sensors](#) > **[optical sensors](#)**

[engineering and technology](#) > [electrical engineering](#), [electronic engineering](#), [information engineering](#) > [electronic engineering](#) > [robotics](#) > [autonomous robots](#) > **[drones](#)**

[agricultural sciences](#) > [agriculture, forestry, and fisheries](#) > **[agriculture](#)**



Keywords ⓘ

[Micro aerial vehicles](#)

[drones](#)

[computer vision](#)

[event cameras](#)

[machine learning](#)

Programme(s) ⓘ

H2020-EU.1.1. - EXCELLENT SCIENCE - European Research Council (ERC) MAIN PROGRAMME

[See all projects funded under this programme](#)

Topic(s) ⓘ

[ERC-2019-COG - ERC Consolidator Grant](#) 

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Funding Scheme

ERC-COG - Consolidator Grant

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Call for proposal

[ERC-2019-COG](#) 

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Host institution



UNIVERSITAT ZURICH

Net EU contribution 

€ 2 000 000,00

Total cost 

€ 2 000 000,00

Address

RAMISTRASSE 71

8006 Zurich

Switzerland 

Region

Schweiz/Suisse/Svizzera > Zürich > Zürich

Activity type

Higher or Secondary Education Establishments

Links

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Beneficiaries (1)



UNIVERSITÄT ZÜRICH

Switzerland

Net EU contribution 

€ 2 000 000,00

Address

RAMISTRASSE 71
8006 Zurich 

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